DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH, LUCKNOW

EVALUATION SCHEME & SYLLABUS

FOR

B. TECH. FOURTH YEAR

CHEMICAL ENGINEERING

AS PER

AICTE MODEL CURRICULUM
[Effective from the Session: 2021-22]

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
B.TECH IV YEAR VII SEMESTER CHEMICAL ENGINEERING

SEMESTER- VII		SESSION 2021-22											
Sl. No	**Subject Codes**	**Subject**	**Periods**			**Evaluation Scheme**				**End Semester**		**Total**	**Credit**
			L	**T**	**P**	**CT**	**TA**	**Total**	**PS**	**TE**	**PE**		
1	KHU701/ KHU702	HSMC -1 #/ HSMC-2 #	3	0	0	30	20	50		100		150	3
2	KCH 071-074	Departmental Elective-IV	3	0	0	30	20	50		100		150	3
3	KCH 075-078	Departmental Elective-V	3	0	0	30	20	50		100		150	3
4		Open Elective-II	3	0	0	30	20	50		100		150	3
5	KCH 751	LAB-1	0	0	2				25		25	50	1
6	KCH 752	Mini Project or Internship Assessment*	0	0	2				50			50	1
7	KCH 753	Project I	0	0	8				50		100	150	4
8		MOOCs (Essential for Hons. Degree)											
		Total										**850**	**18**

*The Mini Project or internship (4 weeks) conducted during summer break after VI semester and will be assessed during VII semester.

DEPARTMENTAL ELECTIVE - IV

KCH-071 Energy Engineering & Management
KCH-072 Plant Design and Economics
KCH-073 Petroleum Refining Technology
KCH-074 Industrial Pollution Abatement and Waste Management

DEPARTMENTAL ELECTIVE - V

KCH-075 Fertilizer Technology
KCH-076 Process Utilities and Safety
KCH-077 Advanced Separation Techniques
KCH-078 Nanotechnology Applied To Chemical Engineering

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
B.TECH IV YEAR VIII SEMESTER CHEMICAL ENGINEERING

SEMESTER- VIII **SESSION 2021-22**

Sl. No	Subject Codes	Subject	Periods			Evaluation Scheme				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1	KHU801 KHU802	HSMC-2#/HSMC-1#	3	0	0	30	20	50		100		150	3
2		Open Elective-III	3	0	0	30	20	50		100		150	3
3		Open Elective-IV	3	0	0	30	20	50		100		150	3
4	KCH 851	Project	0	0	18				100		300	400	9
5		MOOCs (Essential for Hons. Degree)											
		Total										**850**	**18**

SUBJECT CODE: KCH 071	COURSE TITLE: ENERGY ENGINEERING & MANAGEMENT
EXAM DURATION: 3 Hrs	SEMESTER: VII (ODD)
L: T:P::3:0:0 CREDITS: 3	PREREQUISITE: NIL

COURSE OBJECTIVES:

The objective of this course is to

- Create awareness in every engineering graduate about the significance of sustainable utilization of energy.
- Understand the effect of excessive utilization of fossil fuel on the environment.
- Enhance the knowledge on present and future energy challenges in our society.

COURSE OUTCOME : Students completing the course will be able to

- Provide an overview of renewable and non-renewable energy resources scenarios.
- Perform energy audits in various unit operations.
- Able to understand the principles and technologies involved in alternate sources of energy
- Explore the energy conservation opportunities in chemical process utilities
- Study the case studies of energy conservation in chemical process industries

REFERENCE BOOKS

S. NO.	Name Of Authors/Books/Publishers	Year Of Publication/Reprint
1.	Rai, G.D., Non, conventional Energy Sources, Khanna Publishers, New Delhi,	1984.
2.	Dunlap, R. A., Sustainable Energy‖, SI Edition, Cengage Learning India Private Limited, Delhi,	2015.
3.	Kothari, D. P., Singal, K. C. and Ranjan, R., —Renewable Energy Sources and Emerging Technologies‖, 2e, PHI Learning Private Limited, Delhi,.	2011.
4.	Sukhatme, S. and Nayak, J., —Solar Energy: Principles of Thermal Collection and Storage‖, 4e, Tata McGraw-Hill Publishing Company Limited, New Delhi, McGraw Hill.	2017
5	Rao, S. and Parulekar, B.B., Energy Technology, Khanna Publishers.	2005

COURSE DETAIL

Units	Contents	Lecture Hours
I	**Introduction** Renewable and non-renewable energy,Energy Needs of Growing Economy, Long Term Energy Scenario, Energy and Environment: Climate Change, Energy Security, Energy Conservation and its Importance, Energy Strategy for the Future, Kyoto Protocol, Global warming, Fossil and Processed Fuel.	8

II	**Alternate source of energy** Biomass and its conversion roots to gaseous and liquid fuels, Solar Energy : Photo thermal and photovoltaic conversion and utilization methods , solar water heating , cooking , drying and its use for other industrial processes , solar cells their material and mode of operation . Direct and indirect methods solar energy storage, sensible heat and latent heat storage materials, solar ponds. Geothermal energy sources, energy from wind and tides.	**8**
III	**Energy conservation** Energy conservation opportunities in chemical process utilities - Steam systems, Compressed air systems, Insulation, Cogeneration, Cogeneration systems, Waste heat recovery: Sources of waste heat, Feasibility of waste heat recovery, Types of heat recovery equipment.	**8**
IV	**Energy audit and Energy Management** Objectives of energy audit, Energy audit Methodology, Types of energy audit-Preliminary and detailed, The objective of energymanagement,Energy policy & energy planning.	**8**
V	**Case Studies** Energy conservation in the alcohol industry. Energy conservation in the fertilizer industry Energy conservation in the pulps & paper industry. Energy conservation in different units of petroleum industries Energy conservation in various units of tanneries	**8**
	TOTAL	**40**

SUBJECT CODE: **KCH 072**	COURSE TITLE**: PLANT DESIGN AND ECONOMICS**
EXAMINATION DURATION: **3 Hrs.**	SEMESTER: **VII (ODD)**
L:T:P::3:0:0 CREDITS :3	PRE-REQUISITE: **KCH 302, KCH303, KCH 402, KCH 501, KCH 503, KCH 603**

OBJECTIVE

- The objective of the course is to expose students to basic concepts in engineering economics, plant design, safety features and its importance to chemical engineering

COURSE OUTCOME: On completion of this course, the students will be able to

- To learn basic economic concept, to understand and apply this concepts in the project works undertaken and to chemical engineering situation by solving problem
- Carry out the primary techno-economic feasibility of project.
- Select appropriate process for a project.
- Differentiate the equipment and able to prepare specification sheet
- Understand piping and instrumentation diagram
- Evaluate the project cost including capital investment, product cost, breakeven point, depreciation cost for equipment and the total project cost.

REFERENCE BOOKS:

S. No	Name of Authors / Books / Publishers	Year Of Publication/Reprint
1.	M.S. Peters and Timmerhaus, "Plant design and Economics for Chemical Engineers",McGraw Hill, Inc. 4th Edition.	1991
2.	F.C. Vibrandt and C.E. Dryden, "Chemical Engineering Plant Design", McGraw Hill, Fifth Edition.	1959
3.	R. K. Sinnott, Chemical Engineering Design, Volume 6, Elsevier Butterworth-Heinemann, 4th Edition.	2005
4.	V. V. Mahajani, S. M. Mokashi, Chemical Project Economics, Macmillan Publishers India Ltd., 1st Edition, 2005.	2005
5.	O. P. Khanna, Industrial Engineering and Management, Dhanpat Rai Publications, 17th Edition, 2010.	2010
6.	Coulson & Richardson's Chemical Engineering Volume 6, Butterworth-Heinemann, 3rd Edition.	1999

COURSE DETAILS:

Units	Contents	Lecture Hours
I	**Introduction:** Basic considerations in chemical engineering plant design, optimization and feasibility of plant design. **Plant location and layout:** Factors affecting plant location, factors in planning layouts, principles of plant layout, use of scale models.	8
II	**Process design aspects:** Selection of process-factors affecting process selection. Types of project design, Importance of Laboratory development pilot plant, safety factors, types of flow diagrams. **Selection of process equipment:** Standard versus special equipment-material of construction for process equipment, selection criteria, and specification sheets.	8
III	**Cost estimation:** Cash flow and cumulative cash position for industrial operations, factors affecting estimation of investment and production cost, breakeven point and its significance, total capital investment, fixed and working capital investment & their estimations, type of estimates, cost indexes, method for estimating capital investment. **Estimation of total product cost:** Estimation of total product cost: manufacturing cost, general expenses, Manufacturing cost: direct production cost, fixed charges, plant overhead cost.	10
IV	**Depreciation:** Types of depreciation, Method for determining depreciation: straight line method, decline balance method, sum of the year digit method, shrinking fund method etc, single unit and group depreciation, adjustment of depreciation account, evaluation of depreciation methods	7
V	**Profitability, alternative investments and replacement:** Methods for profitability evaluation, Evaluation of Break Even Point, % rate of return, Practical factors in alternative investment and replacement Studies.	7
	TOTAL	**40**

SUBJECT CODE: KCH 073	COURSE TITLE: PETROLEUM REFINING TECHNOLOGY
EXAM DURATION: 3 Hrs	SEMESTER: VII (ODD)
L:T:P::3:0:0 CREDITS :3	PREREQUISITE: NIL

Course Objectives: The objectives of this course is to

- Study the petroleum exploration and Classification of crude oils
- Understand the essential products characteristics.
- Develop skills in drawing neat flow diagrams of different petroleum refining processes.
- Understand the final product treating operations.
- Identify important testing methods for petroleum products.

Course Outcome : Students completing the course will be able to

- Describe the process of crude oils production & refining and Characteristics of crude oils
- Understand the various quality Control parameters of Petroleum Products
- Describe the physical properties of petroleum products and thermal conversion of petroleum products
- Understand the process involved in catalytic conversion
- Demonstrate the different methods available for lube oil manufacturing process

REFERENCE BOOKS

S. NO.	Name of Authors/Books/Publishers	Year of Publication/Reprint
1	Robert J. Mayer, "Handbook of Petroleum Refining Processing", McGraw-Hill Education	2015
2	Nelson, W.L., Petroleum Refining Engineering, McGraw Hill	1969
3	Mohamed A. Fahim, "Fundamentals of Petroleum Refining", Elsevier science	2009
4	Bhaskara Rao B., "Modern petroleum refining processes", oxford & IBH publishing	2018
5	Speight James G. "Handbook of Petroleum Refining" ,Taylor & Francis Ltd	2016
6	Raseev S ,"Thermal and Catalytic Processes In Petroleum Refining", Taylor & Francis Ltd.	2003

COURSE DETAIL

Units	Contents	Lecture Hours
I	**Introduction** Petroleum Exploration: Production and Refining of Crude oils, Crude oils: Chemistry and composition (Characteristics and constituents of crude oils, classification of crude oils)	8
II	**Thermal Conversion Process and Petroleum Products** Crude Oil Distillation Desalting of crude oils, Atmospheric distillation of crude oil, Vacuum distillation of atmospheric residue. Composition, Properties & Specification of LPG, Naphthas, motor spirit, Kerosine, Aviation Turbine Fuels, Diesel Fuels, Fuel Oils, Petroleum Hydrocarbon Solvents, Lubricating oils (automotive engine oils, industrial lubricating oils electrical insulating oils, Jute Batching oils, white oils, steam turbine oils, metalworking oils, etc.) Petroleum Waxes Bitumen's, Petroleum coke. Thermal Conversion Process, Thermal Cracking Reactions, Thermal Cracking, Visbreaking, (Conventional Visbreaking and Soaker Visbreaking) Coking (Delayed Coking, Fluid Coking, Flexicoking), Calcination of Green Coke.	8
III	**Catalytic Conversion Process** Fluid catalytic cracking; Catalytic reforming; Hydrocracking catalyticAlkylation, Catalytic Isomerization; Catalytic Polymerization. Finishing Process Hydrogen sulfide removal processes; Sulphur conversion processes; Sweetening processes (Caustic treatment, Solutizer process; Doctor treating process; Copper chloride sweetening,; Hypochlorite sweetening; Air and inhibitor treating process; Merox processes; Sulphuric acid treatment; Clay treatment); Solvent extraction processes (Edeleanu process, Udex process, Sulfolane process), Hydrotreating processes.	8
IV	**Quality Control of Petroleum Products** Classification of laboratory tests, distillation, vapour pressure, flash and fire points, octane number, performance number, cetane number, aniline point, viscosity index, calorific value, smoke point, char value, viscosity, viscosity index, penetration tests, cloud and pour points, drop point of grease, melting and settling points of wax, softening point of Bitumen, induction period of gasoline, the thermal stability of jet fuels, gum content, Total Sulphur, Acidity and Alkalinity, Copper Strip Corrosion Test, Silver – Strip Corrosion Test for ATF, Ash, Carbon Residue (Conradson method, Ramsbottom method) Colour, Density and Specific gravity, Refractive index of hydrocarbon liquids.	8
V	**Lube Oil Manufacturing Process** Evaluation of crude oils for lube oil base stocks, Vacuum distillation, Solvent deasphalting Solvent extraction of lube oil fractions (Furfural, NMP and Phenol),Solvent dewaxing, Hydrofinshing, Manufacture of petroleum waxes (Wax sweating,Solventdeoiling) Manufacture of Bitumen's Selection of crude oil, Methods of manufacture of bitumens, (Distillation, Solvent precipitation, Air blowing).	8
	TOTAL	**40**

SUBJECT CODE: **KCH 074**	COURSE TITLE**: Industrial Pollution Abatement & Waste Management**
EXAMINATION DURATION: **3 Hrs.**	SEMESTER: **VII (ODD)**
L:T:P:: 3:0:0 CREDITS: 03	PRE-REQUISITE: **Mass Transfer, Chemical Engg. Fluid Mechanics, Heat Transfer, and CRE I**

OBJECTIVE:

- The objective of the course is to impart knowledge on sources and characteristics of various industrial wastes and strategies for its prevention and control

COURSE OUTCOME: On completion of this course, the students will be able to:

- Develop an insight into the pollution from major industries including the sources and characteristics of pollutants
- To plan minimization of industrial wastes
- To design facilities for the processing and reclamation of industrial waste water

REFERENCE BOOKS:

S. No	Name of Authors / Books / Publishers	Year of Publication/Reprint
1.	Metcalf & Eddy, "Wastewater Engineering - Treatment and Reuse", Revised by G. Tchobanoglous, F. L. Burton, and H. D. Stensel, 4th edition. Tata McGraw-Hill.	2003
2.	Mahajan S. P., Pollution control in process industries, Tata McGraw-Hill.	1985
3.	Peavy H.S., Rowe D.R. and Tchobanoglous G., Environmental Engineering, McGraw- Hill Inc.	1985
4.	Kreith F. and Tchobanoglous G., "Handbook of Solid Waste Management", 2nd Ed., Mc Graw Hill Education.	2002
5.	Pichtel J., "Waste Management Practices: Municipal, Hazardous and Industrial", CRC Press.	2005
6.	Shen Thomas T., "Industrial Pollution Prevention", Publisher: Springer, Berlin, Heidelberg, edition 1.	1995
7.	Freeman H.M., "Industrial Pollution Prevention Hand Book", McGraw Hill Inc., New Delhi.	1995
8.	Industrial Wastewater Management, Treatment and Disposal",(WEF – MOP – FD3) McGraw Hill, 3rd Edition	2008
9.	E-sources: https://pubsaskdev.blob.core.windows.net/pubsask-prod/81952/e10-22.pdf	2010

COURSE DETAILS:

Units	Contents	Lecture Hours
I	**Introduction:** Industrial Pollution and types of pollution from chemical process industries, Characterization of emission and effluents, Global consideration of environmental pollution. Effects of industrial effluents on streams, sewer, land, sewage treatment plants and human health, Standards for liquid effluents from chemical process industries, air quality, nuclear radiation emission, noise emission.	8
II	**Air Pollution Control:** Air pollutants classification, Equipment for controlling particulate and gaseous pollutants, lapse rate, atmospheric stability, Dispersion models, Plume behavior, Stack design, Design of gravity settling chamber, cyclones, electrostatic precipitator, fabric filters and absorbers, Air pollution control for petroleum refineries and cement plants. Air Act 1981.	8
III	**Water Pollution Control:** Waste water characteristics, Primary, secondary and tertiary treatments for wastewater, Anaerobic and aerobic treatment biochemical kinetics, Design of trickling filter, activated sludge systems, ponds and lagoons and aeration systems, Water pollution control for petroleum refineries, fertilizer industry, pulp and paper industry. Water Act 1974.	8
IV	**Solid Waste Management**: Characterization of solid wastes-hazardous and non-hazardous wastes, Waste disposal and management laws and guidelines, Non-hazardous industrial wastes treatment, disposal, utilization and management, Value-extraction from the wastes, Handling, storage and disposal of hazardous wastes, Waste disposal for nuclear power plants. The Solid Wastes Management Rules 2016.	8
V	**Pollution Prevention:** Process modification, Alternative raw material, Recovery of by product from industrial emission/effluents, Recycle and reuse of waste, Energy recovery and waste utilization, Material and energy balance for pollution minimization, Water minimization, Fugitive emission/effluents and leakages and their control-housekeeping and maintenance. EnvironmentalProtectionAct1986. The Environmental Management and Protection Act, 2010. Central Pollution Control Board Guidelines.	8
	TOTAL	**40**

SUBJECT CODE: KCH 075	COURSE TITLE: FERTILIZER TECHNOLOGY
EXAM DURATION: 3 HOURS	SEMESTER: VII (ODD)
L: T: P :: 3 : 0 : 0 CREDITS: 3	PREREQUISITE: BASICS OF CHEMICAL PROCESS INDUSTRIES AND CHEMICAL TECHNOLOGY

OBJECTIVE:

- To provide comprehensive understanding of essential ingradients of fertilizers and the production processes of different fertilizers.

COURSE OUTCOME: On successful completion of the course, the student will be able to:

- Use reactions and unit operations steps in manufacturing of various fertilizers.
- Select the relevant manufacturing process for different fertilizers.
- Identify engineering problems in fertilizer manufacturing.
- Select appropriate fertilizer on the basis of properties.

REFERENCE BOOKS:

S. NO.	NAME OF AUTHORS/BOOKS/PUBLISHERS	Year Of Publication/ Reprint
1.	M. Gopala Rao & Marshall Sittig, Dryden's Outlines of Chemical Technology, East-West Press, 3rd Edition, New Delhi.	2015
2.	Austin G. T, Shreve's Chemical Process Industries, 5th edition, Mc. Graw Hill Publications.	2017
3.	Pandey & Shukla, Chemical Technology, Volume I & II, 2nd Edition, Vani Books Company.	2018
4.	John J. Mortvedt, Roy H. Follett, Larry S. Murphy, Fertilizer Technology and Application, Meister Publishing Company.	1999
5.	Slack A.V., Chemistry & Technology of Fertilizers, Interscience, New York.	1967

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	**Introduction to fertilizers:** Introduction to Plant nutrients, Fertilizer specifications, Terminology and Definitions, Classifications of Soil nutrients, Fundamentals of Soil Nitrogen, Soil Phosphorus, Soil Potassium and Soil Sulfur.	7

II	**Nitrogenous Fertilizers:**Methods of production, characteristics, storage and handling specificationsof Ammonium Nitrate, Calcium Ammonium Nitrate, Ammonium Chloride, Ammonium Sulphate, Urea.	**8**
III	**Phosphatic Fertilizers:**Raw materials: Phosphate rock, Sulphur pyrites, Sulphuric acid and Phosphoric acid, Methods of production, characteristics and specifications of Single Super Phosphate, Triple Super phosphate, Ammonium Phosphates: SAP, DAP.	**8**
IV	**Potassic Fertilizers:** Methods of production, specification, characteristics of Potassium Sulphate,Potassium Chloride, Potassium Nitrate Compound Fertilizers, NPK Fertilizers, Complex Fertilizers.	**8**
V	**Fertilizers and Environment:** Environmental issues related to the use of fertilizers, impact of fertilizers on the environment, Solid, Liquid and Gaseous pollution from fertilizer industries, Bio fertilizers, Green fertilizers.	**9**
	TOTAL	**40**

SUBJECT CODE: KCH 076	COURSE TITLE: PROCESS UTILITIES AND SAFETY
EXAM DURATION: 3 HOURS	SEMESTER: VII (ODD)
L: T: P :: 3 : 0 : 0 CREDITS: 3	PREREQUISITE: NIL

OBJECTIVE:

- To gain knowledge about different process utilities used in the chemical process industry
- To understand issues related to hazards & safety in chemical industry.

COURSE OUTCOME: On successful completion of the course, the student will be able to:

- Calculate the requirements of water and air and their applications as utilities.
- Calculate the steam requirement and its applications as utility.
- Evaluate and apply the various risk assessment methods in industries.
- Do the hazard analysis for different industries using HAZOP.

REFERENCE BOOKS:

S. NO.	NAME OF AUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1.	Nordell, Eskel, "Water Treatment for Industrial and Other Uses", Reinhold Publishing Corporation, New York.	1961
2.	Crowl, D.A. &Louvar, J.F.. "Chemical Process Safety: Fundamentals with Applications". New Jersey: Prentice-Hall.	1989
3.	Goodall, P. M., "The Efficient Use Of Steam" IPC Science and Technology.	1980
4.	Lees, F. P., "Loss Prevention in Process Industries 3 volume set"Butterwort -Heinemann,Oxford.	1996

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	**Process Utilities** Various process utilities, their role and importance in chemical plants.Water, compressed air, steam, heat transfer fluids, vacuum, refrigeration, venting, flaring and pollution abating **Water Utilization** Sources of water and their characteristics; Treatment storage and distribution of water; water for use in boilers, cooling purposes, drinking and process; Reuse and conservation of water; Water resource management.	8
	Steam Generation and Utilization Steam generation and its application in chemical process plants, distribution and	8

II	utilization;Design of efficient steam heating systems; Types of boilers and their operation, Selection and sizing of boilers; waste heat boilers, Steam generation by utilizing process waste heat using thermic fluids Steam economy, Steam condensers and condensateutilization Expansion joints, flash tank design, steam traps their characteristics, selection andapplication, waste heat utilization; Lagging, selection and thickness.	
III	**Compressors, blowers and Vacuum Pumps** Compressors, blowers and vacuum pumps and their performance characteristics; Methods ofdeveloping vacuum and their limitations, material handling under vacuum, Piping systems;Lubrication and oil removal in compressors and pumps.Air filters,Air and gas leakage. Inert gas systems, compressed air for process,Instrument air, Pneumatic control and conveying, Refrigeration and HVAC fundamentals. **Insulation** Importance of insulation for meeting the process requirement, insulation materials and theireffect on various material of equipment piping, fitting and valves etc. insulation for highintermediate, low and subzero temperatures, including cryogenic insulation.	**8**
IV	**Elements of Safety** Elements of safety, safety and site selection; Plant layout and unit plot planning; Definition of riskand hazard, Identification and assessment of the hazards, distinction between hazards and risk,Industrial hygiene, toxicological studies, Hazard operability (HAZOP) hazard analysis (HAZAN); Assessment of the risk, fault tree, eventtree, scope of risk assessment; Control of hazards, controlling toxic chemicals and controllingflammable materials. **Prevention of losses** Prevention of losses, Pressure relief, Prevention and control of fire and explosion risk, Provision of firefightingequipment, Technology selectionand transfer, Choosing the right process.	**8**
V	**Control of Process** Control of process, Prevention of hazardous deviation in process variables, e.g. pressure,temperature flow by provision of automatic control systems- interlocks, alarms, trips together withgood operating practices and management. **Safety Regulations** Regulations and legislation, Risk management routines and tackling disaster, Safety Procedures and Designs Process Safety Hierarchy, Process Safety Strategies, Managing Safety, Safety Reviews and Accident Investigations, Designs for Process Safety, Inherently Safer Designs.	**8**
		40

SUBJECT CODE: KCH 077	COURSE TITLE: ADVANCED SEPARATION TECHNIQUES
EXAM DURATION: 3 HOURS	SEMESTER: VII (ODD)
L: T: P :: 3 : 0 : 0 CREDITS: 3	PREREQUISITE: NIL

OBJECTIVE:

- To understand the governing mechanisms and driving forces of various advanced separation processes
- To be able to understand process variables and calculations

COURSE OUTCOME: On successful completion of the course, the student will be able to:

- Understand the principles of separation processes.
- Utilize mass transfer concepts for separation systems.
- Discuss and explain the basics of separation process and its application
- Perform design calculations for advanced separation processes.
- Analyze and use the concept of modern separation techniques in various applications
- Ability to Select appropriate separation technique for intended problem.

REFERENCE BOOKS:

S. NO.	NAME OF AUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1.	Seader J.D., Ernet J. Henlay, and Keith, D., "Separation Process Principles",John Wiley & Sons, Inc.	2010
2.	King, C.J., Separation Processes, Tata McGraw - Hill Publishing Co., Ltd.	1982
3.	Osadar, V., and Nakagawa, I.,Membrance Science and Technology, Marcel Dekkar	1992
4.	Schoew, H.M., New Chemical Engineering Separation Techniques,Interscience Publishers	1972
5.	Geankoplis, C.J. —Transport Processes and Unit Operations‖, 3e, Prentice Hall	2003
6.	R W Rousseau, "Handbook of Separation Process Technology", John Wiley & Sons, Inc.	1987

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	Introduction to separation, Uses and characterization of separation processes, Fundamentals and classification of various types of separation processes, governing mechanisms.	**6**
II	Multiple separation units: Cascades and their types, Need for cascades, cascade arrangement and interstage flows, squared off cascades, Separative duty and potential	**6**
III	Membrane based separation processes: Principles, mechanisms and Classifications of membrane separation processes, Membrane characterizationand various membrane modules used in membrane separation processes, flow pattern, Gas permeation through polymer membranes, Liquid membrane sepation process, application & advantages of membrane separation processes.	**10**
IV	Concept & working principal and applications of various separation processes: Dialysis, Reverse osmosis, Microfiltration, Ultrafiltration, Pervaporation, Electro-dialysis Effect of Concentration polarization in membrane processes	**8**
V	Chromatographic separation processes: Fundamentals and types of chromatography, Classification and separation mechanism. Molecular sieve separations: classification and applications	**10**
		40

SUBJECT CODE: KCH 078	**COURSE TITLE: NANOTECHNOLOGY APPLIED TO CHEMICAL ENGINEERING**
EXAM DURATION: 3 HOURS	**SEMESTER: VII (ODD)**
L: T: P :: 3 : 0 : 0 CREDITS: 03	**PREREQUISITE: FUNDAMENTAL OF CHEMISTRY AND THERMODYNAMICS**

OBJECTIVE:

- To make the students understand the concept of Nanoparticles.
- To help them understand in broad outline of Nanoscience and Nanotechnology.
- To make them understand the synthesis of nanomaterials and their uses in various applications.

COURSE OUTCOME: On successful completion of the course, the student will be able to:

- Understand the concept and uses of nanoparticles.
- Understand bulk and Nanostructured materials.
- Understand the basic principal of various characterization technique.
- Understand the use of nanoscience and nanotechnology for various applications.
- Understand synthesis of nanomaterial and the difficulties in synthesizing Nano particles
- Select appropriate nanoparticles for intended problem. Work in the field of commercialization of Nano materials.

REFERENCE BOOKS:

S. NO.	NAME OF AUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1.	Introduction to Nanotechnology by Charles P. Poole Jr and. Frank J. Owens, Wiley-Inter science.	2003
2.	Textbook of Nanoscience and Nanotechnology, B.S. Muty, P. Shankar, Baldev Raj, B.B Rath and James Murday, University Press.	2013
3.	Nanoscale Materials in Chemistry Edited by Kenneth J. Klabunde, John Wiley & Sons Inc.	2001
4.	Nanostructures and Nanomaterials: Synthesis, Properties and Applications by G. Cao, Imperial College Press	2004
5.	Nanoscale Science and technology by Robert Kelsall (editor), Ian W. Hamley (co-editor), Mark Geoghegan (co-editor) , John Wiley & Sons Inc.	2005
6.	The Chemistry of Nanomaterials: Synthesis, Properties and Applications by C. N. R. Rao, A. Muller, A. K. Cheetham, WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.	2004

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	**Generic methodologies for nanotechnology:** Introduction, classification and fabrication, Summary of the electronic properties of atoms and solids, Effects and importance of the nanometer length scale, Fabrication methods, Preparation, safety and storage issues, some key examples of nanotechnology.	**6**
II	**Strategies for the scalable synthesis of Nano dimensional materials**: Introduction, Defining Nano dimensional Materials, Potential Uses and The General Methods Available for the Synthesis of Nano-Materials, Precipitative Methods, Reactive Methods in High Boiling Point Solvents, gas-phase synthesis of semiconductor nanoparticles, synthesis in a structured medium, the suitability of such methods for scaling	**10**
III	**Nanoparticles:** Wet chemical synthesis, preparation and properties of iron, platinum, gold, cadmium, silver, copper and nickel nano-particles.	**8**
IV	**Tools to characterize nanomaterials**: X-Ray Diffraction (XRD), Scanning Electron Microscopy, Transmission Electron Microscopy, Atomic Force Microscopy,UV/Visible Spectroscopy, AAS	**8**
V	**Applications of nanomaterials:** Nano catalysts, Water Treatment and the Environment, Paints, Food and Agriculture Industry, Cosmetics and Consumer Goods, Nano Sensor.	**8**
		40

SUBJECT CODE:**KCH 751**	COURSE TITLE: **ENERGY LAB**
EXAMINATION DURATION: 3Hrs	SEMESTER: **VII (ODD)**
L:T: P::0:0: 2 CREDITS: 1	PRE-REQUISITE: **NIL**

OBJECTIVE:

- To make students understand the basic fundamental principles of energy engineering by performing different experiments
- To make them correlate theory and the practical process by experimentation.

LAB OUTCOME: On successful completion of the course, the student will be able to

- Analyze the data on the properties of solid fuels.
- Analyze the data on the properties of liquid fuels.
- Write technical reports effectively with proper conclusions.

REFERENCE BOOKS:

S. NO.	Name of Authors/Books/Publishers	Year of Publication/Reprint
1.	Rai, G.D., Non, conventional Energy Sources, Khanna Publishers, New Delhi.	1984.
2.	Dunlap, R. A., Sustainable Energy‖, SI Edition, Cengage LearningIndia Private Limited, Delhi,.	2015.
3.	Kothari, D. P., Singal, K. C. and Ranjan, R., —Renewable EnergySources and Emerging Technologies‖, 2e, PHI Learning Private Limited, Delhi,.	2011.
4.	Sukhatme, S. and Nayak, J., Solar Energy: Principles of ThermalCollection and Storage‖, 4e, Tata McGraw-Hill Publishing CompanyLimited, New Delhi,McGraw Hill.	2017
5	Rao, S. and Parulekar, B.B., Energy Technology, Khanna Publishers.	2005

LIST OF EXPERIMENTS

S. No.	EXPERIMENT
1	To determine flash and fire point – Pensky Martin apparatus
2	To determine the viscosity of oil – Saybolt viscometer
3	To determine the Calorific Value of given Solid Fuel
4	To determine the Calorific Value of given Liquid Fuel
5	To determine the Moisture Content in a given coal sample.
6	To determine the Volatile Content in a given coal sample.
7	To determine the Ash Content in a given solid fuel sample.
8	To determine the cetane number and cetane index of diesel
9	To determine the smoke point of liquid fuels
10	To determine the octane number of gasoline products